Spotting Untapped Investment Opportunities in Nashville via Machine Learning & Real-Time Sentiment

Hello everyone, we are Team Pluto. Today we present our scalable analytics platform focused on the short-term rental market in Nashville, Tennessee.

Team Pluto
Presenter

Unified Data & Cloud Architecture

Integrating Yelp Academic Dataset and Inside Airbnb on AWS for scalable batch and stream analytics

Primary data sources: Yelp Academic Dataset; Inside Airbnb

Yelp captures local business activity while Inside Airbnb details rental listings for combined analysis.

Cloud deployment: AWS EC2 + Docker

Services run in Docker on EC2 for scalable, modular orchestration.

Storage & ingestion: AWS S3 buckets

Raw data stored and ingested from S3 for durable, efficient access.

Workflows & integration: Batch and streaming pipelines

Used Kafka for streaming, Machine Learning for nalysis and streamlit for visualization.

Data Acquisition & Cleaning Challenges

Key PySpark steps to convert messy Airbnb and Yelp sources into analytics-ready Parquet

Data Ingestion and Processing: Acquired Yelp business data and Airbnb listings; large files downloaded to the project root and ingested via PySpark for scalable processing

Kafka for Real-time Data: Yelp review data was streamed in real-time using a Kafka producer (yelp_producer.py) feeding data into the yelp-reviews topic.

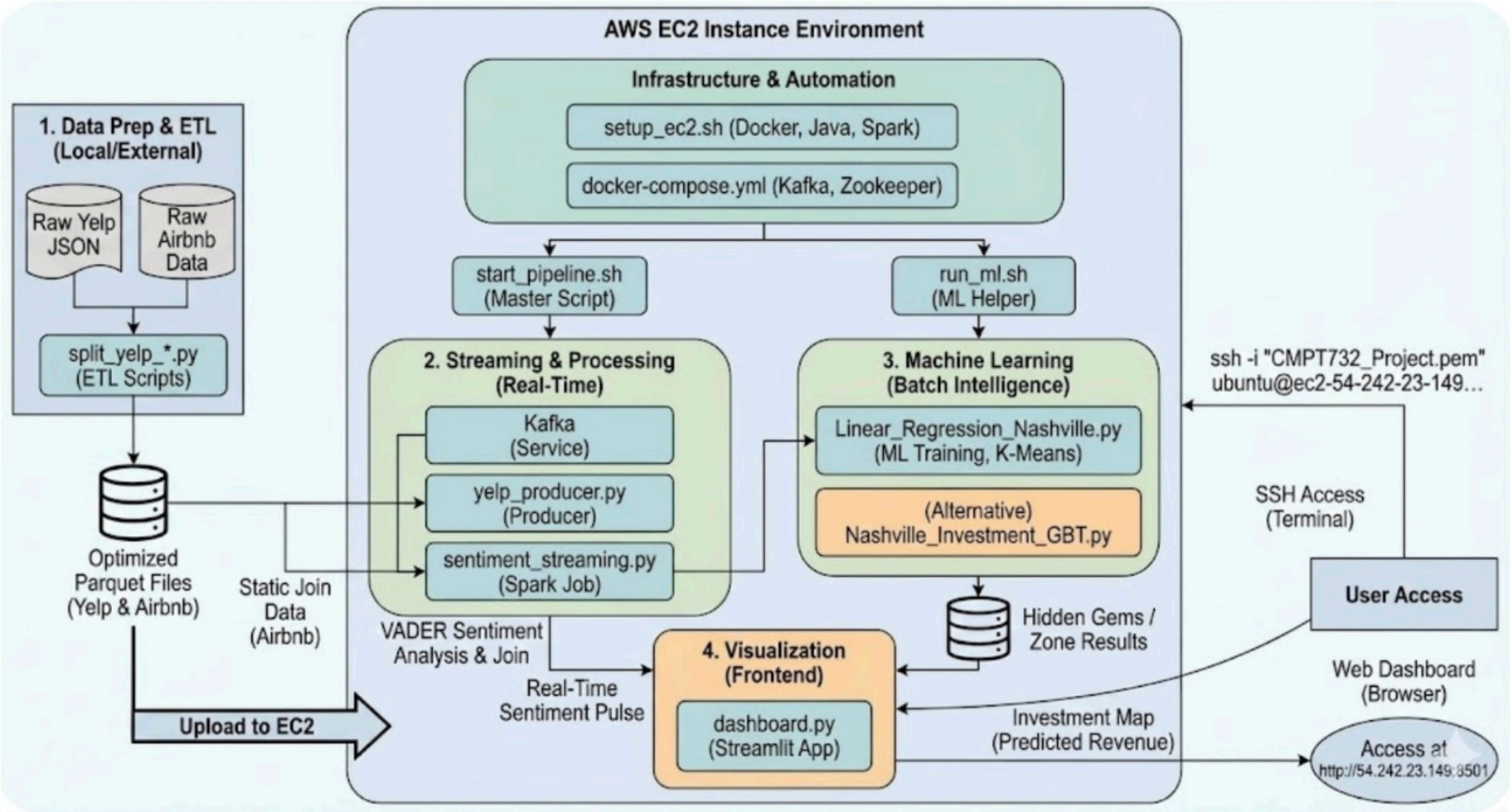
Spark Streaming for Real-time Analysis: A Spark Streaming job (sentiment_streaming.py) consumed the Kafka stream, joined it with Airbnb data, and performed real-time sentiment analysis using VADER. The job also handled checkpointing, windowing, and schema enforcement for robust stream processing.

ETL Scripts for Pre-processing: A suite of ETL scripts (split_yelp_*.py) were used to pre-process the massive raw Yelp JSON dataset, converting it into optimized Parquet files partitioned by city.

Data Cleaning & Preparation: The machine learning script (Linear_Regression_Nashville.py) loaded datasets, executed cleaning and normalization steps, clustered Nashville into 35 zones using K-Means for spatial feature engineering, and prepared feature sets before training the regression model.

Automation of Data Pipeline: A master script (start_pipeline.sh) automated the entire pipeline: it launches Zookeeper & Kafka, starts the Kafka producer, provisions required Parquet sources, and initiates the Spark Streaming job with checkpointing and logging for reliable end-to-end data acquisition and processing.

Project Set-up and Flow



Methodology & Modeling Approach

Geographic clustering, revenue modeling, and real-time sentiment fusion for Nashville

Geospatial Clustering

Applied K-Means to Yelp and Airbnb coordinates; Elbow Method selected 35 micro-neighborhoods to reconcile inconsistent names.

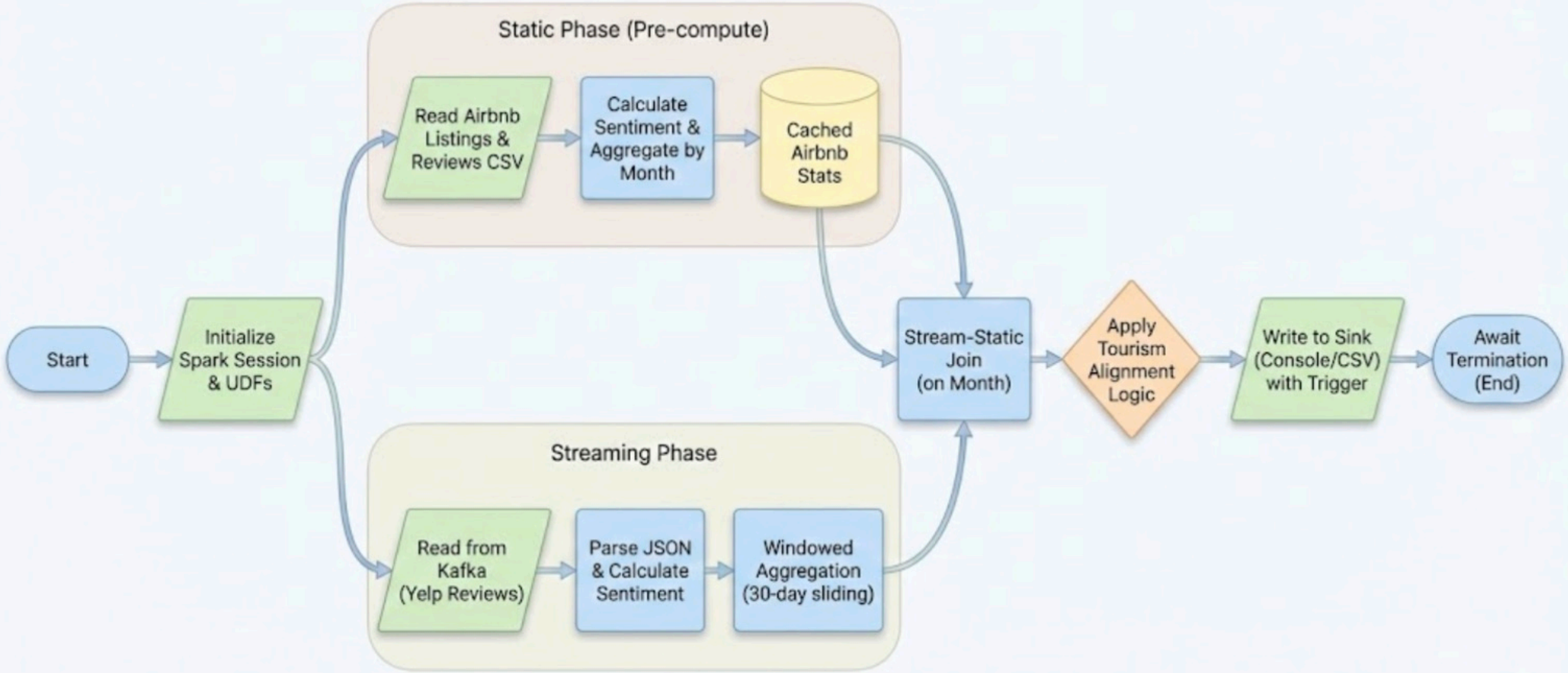
Revenue Modeling

Linear Regression predicted expected revenue per zone using commercial popularity metrics such as total businesses and check-in volume.

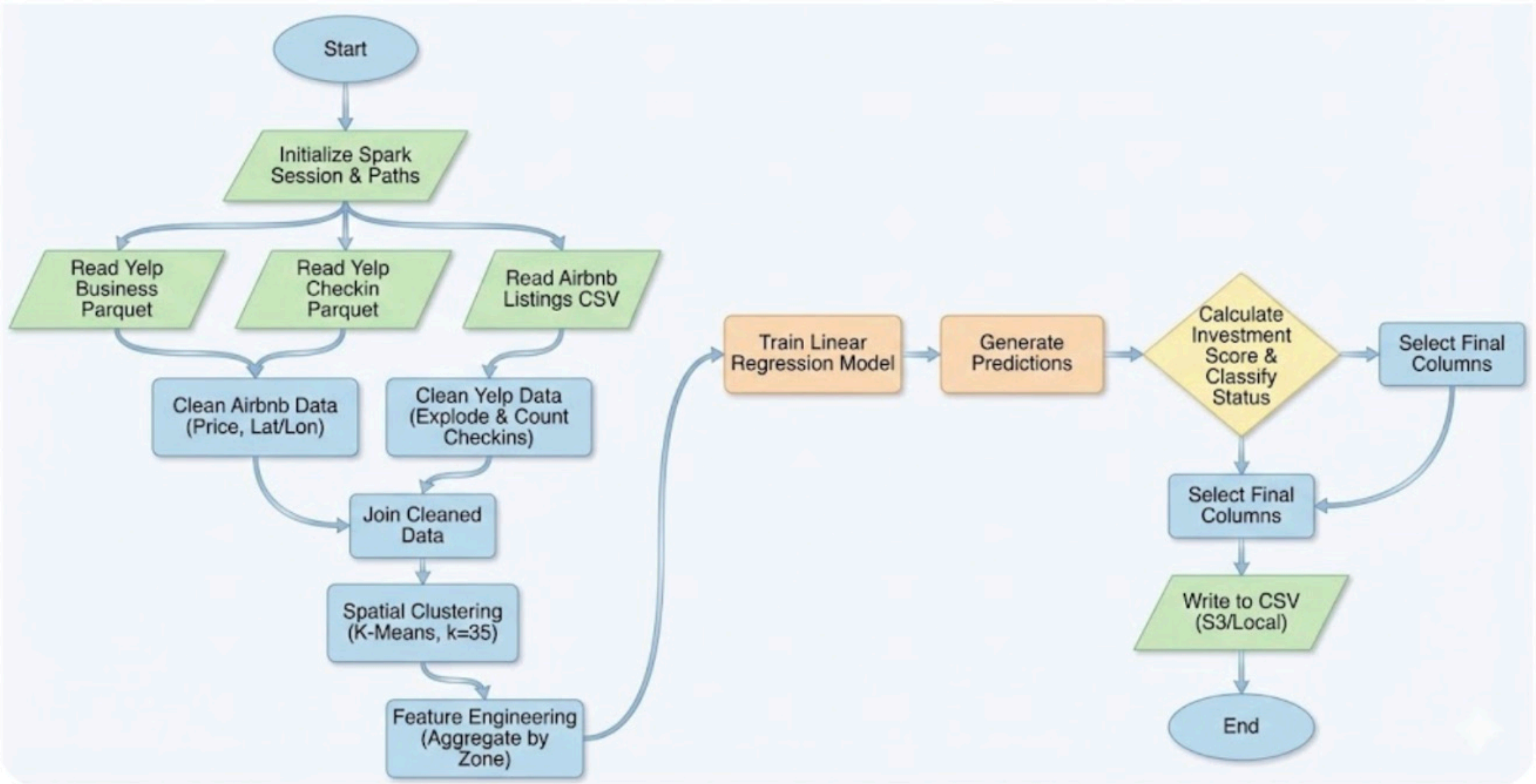
Sentiment Streaming

Simulated real-time Yelp review streaming with Apache Kafka; reviews cleaned with Regex and scored using VADER sentiment analysis.

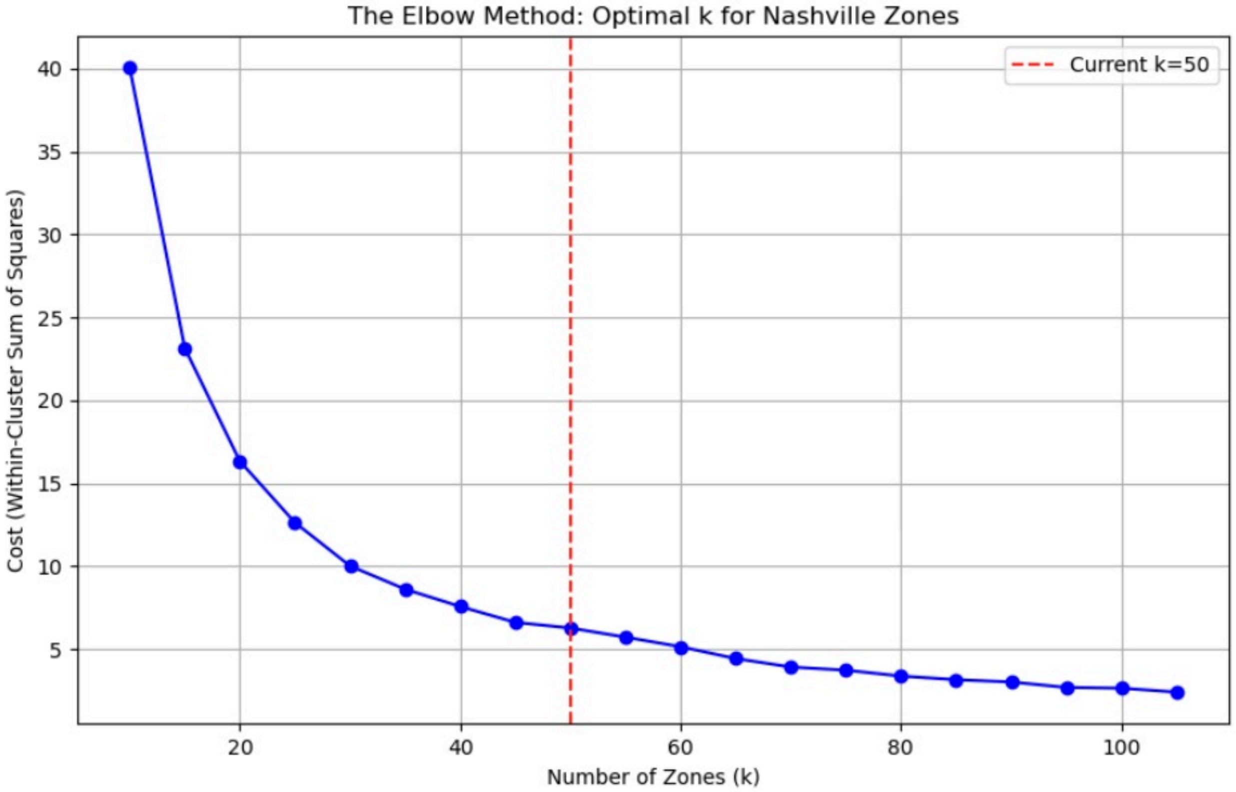
Sentiment Analysis



Investment Analysis



Optimal Zones for Clustering



Challenges & Limitations

Key technical hurdles and pragmatic mitigation for temporal gaps

API Constraints & Costs: Limited access to live Yelp data due to rate limits and high costs hindered meaningful sentiment analysis.

Data Integration Challenges: Mismatches between Yelp and Airbnb records and inconsistent neighbourhood boundaries made geospatial mapping complex and computationally expensive; aligned temporal windows, boundary harmonization, and precomputed spatial joins can mitigate this.

Strategic Pivot to Nashville: To address data scarcity in Vancouver, we shifted focus to Nashville to leverage the richer, historical Yelp Open Dataset and achieve better temporal and spatial alignment with Airbnb records.